SMART WORKSPACE RESERVATION AND ENFORCEMENT SYSTEM

TEAM: BYTE INNOVATORS (SMART CAMPUS)

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PROBLEM STATEMENT

During peak hours of examination there is lack of seats in Library due to reserving of seats by leaving personal belongings without presence.This creates unfair occupancy and over crowding in certain areas which cause dissatisfaction for students.So there is need for a smart system to solve this problem .

There is no proper booking mechanism to manage high demand of students leading to overcrowding which results in poor resource management.

Study rooms and shared workspaces often become overcrowded during peak hours due to high demand and lack of proper usage control. Without effective monitoring and enforcement, users may occupy spaces beyond their reserved time or use rooms without booking. This leads to discomfort, reduced productivity, and unfair access for other users seeking space. Additionally, administrators face difficulties in tracking occupancy levels and preventing overcrowding in real time. Therefore, a smart reservation and enforcement system is needed to regulate usage, prevent overcrowding, and ensure a balanced and efficient allocation of study spaces.

Impact of problems

- Library seats are wasted and underutilized, even during peak demand.
- Students lose time searching for available seats, reducing study efficiency.
- Unfair access occurs, as some students occupy seats without being present.
- Certain areas become overcrowded, while others remain unused.
- Conflicts and arguments arise between students over seat ownership.

Feasibility analysis

- The system can be developed using existing technologies such as mobile apps, web platforms, cloud databases, and simple IoT sensors.

Solutions:

Smart Seat Reservation System

- Students can book seats in advance using a mobile or web app.
- Shows real-time availability of seats across the library.

Anti-Hoarding System

- Detects long unattended periods.
- Sends warnings → then releases the seat if no response.

Ideation

- **Problem**: Study rooms often get overcrowded; manual bookings lead to no-shows and misuse.
- **Goal**: Ensure fair access, prevent overcrowding, and optimize space usage.
- Solution: QR code–based booking and check-in system with real-time monitoring.
- **Key Features**:
- Scan QR to book and check-in
- Auto enforcement of booking rules
- Real-time occupancy updates
- Admin dashboard for monitoring and analytics

Start → User scans QR Code → System shows room details & availability → Is room available?

↓ Yes ↓ No

Book Room → Receive Confirmation Show "Room Not Available"

↓

User scans QR Code to Check-in → Did user check-in on time?

↓ Yes ↓ No

Allow Access Cancel Booking → Notify Others

↓

Monitor Room Occupancy → Is room overcrowded or misused?

↓ Yes ↓ No

Send Alert / Enforce Rules Continue Session

↓

Is booking time over?

↓ Yes ↓ No

Auto Check-out Continue Monitoring

↓

Update Room Availability → End

Robin:
A popular workplace scheduling platform used in offices for desk and room booking.
Teem:
Provides tools for reserving meeting rooms and managing office space.

Challenge: Lack of strict enforcement (no-shows, overstaying)

Solution:

Implement QR code–based check-in to validate actual presence.
Set automatic timeouts: if a user doesn’t check in within a grace period, cancel the booking.
Use alerts or penalties for repeated misuse.

System Design

Architecture Components:

User Interface: QR code scanning via mobile camera or web browser.
Backend: Handles booking logic, check-in validation, occupancy monitoring, and notifications.
Admin Dashboard: Real-time analytics, room management, and policy enforcement

User scans QR → System displays room availability → Booking confirmed → Check-in via QR → Real-time monitoring → Auto enforcement → Check-out → Update availability.

Implementation

Steps:

QR Code Generation: Unique QR for each room linking to booking system.
Database Setup: Tables for users, rooms, bookings, occupancy logs.
Notification System: Email or SMS reminders for booking start/end and alerts for violations.

Prototype Progress

Phase 1: QR code scanning and basic booking system (web-based).

Phase 2: Check-in/check-out functionality with QR validation.

Phase 3: Real-time occupancy monitoring using simulated sensor data.

Phase 4: Notifications and automated alerts (pending integration)

Technical Approach

QR Code System: Unique codes per room linked to the backend for booking and check-in.

Real-Time Updates: WebSocket or periodic polling to update room availability.

Enforcement Rules:

No-show → auto-cancel booking

Overstaying → alert or terminate session

Overcrowding → alert admin

Scalability: Modular architecture allows easy addition of more rooms, users, and sensor integration.

Benefits:

Efficient Space Use – Prevents overcrowding and optimizes room occupancy.
Quick & Contactless Booking – Users book and check in by scanning QR codes.
Automatic Enforcement – Prevents double bookings, no-shows, and overstaying.
Real-Time Monitoring – Admins can track room usage instantly.

Limitations

Dependent on Smartphones – Users must have a smartphone to scan QR codes.
Internet Required – System may not work offline; needs network connectivity.
No Real-Time Presence Without Sensors – QR check-in only confirms user scanned the code, not actual room occupancy.
Potential Misuse – Users may share QR codes or fake check-ins.
Limited Enforcement for Crowding – Overcrowding is hard to detect without additional sensors.

Future scope:

IoT Integration – Automatic check-in/out using sensors.
AI Suggestions – Recommend optimal time slots and predict peak usage.
Multi-Building Support – Manage rooms across campuses centrally.
Automated Alerts – Notifications for reminders, overstaying, or conflicts.
Analytics & **Reporting** – Track usage trends for better planning.

Conclusion:

The QR-based reservation system makes booking and check-in easy, efficient, and conflict-free. It optimizes space usage and provides real-time monitoring. With future upgrades, it can become a fully automated smart workspace management system.